

Cool Roof Priority Index

Satellite screening that turns a city heat map into a **ranked worklist of individual rooftops** — with the cooling-per-retrofit **measured from Abu Dhabi's own data**, and the roof material identified from **NASA EMIT hyperspectral imagery over Mussafah itself**.

Team T0050

United Arab Emirates

Mussafah, Abu Dhabi

195 km² industrial district

Theme

Urban expansion, land-use change & heat risk

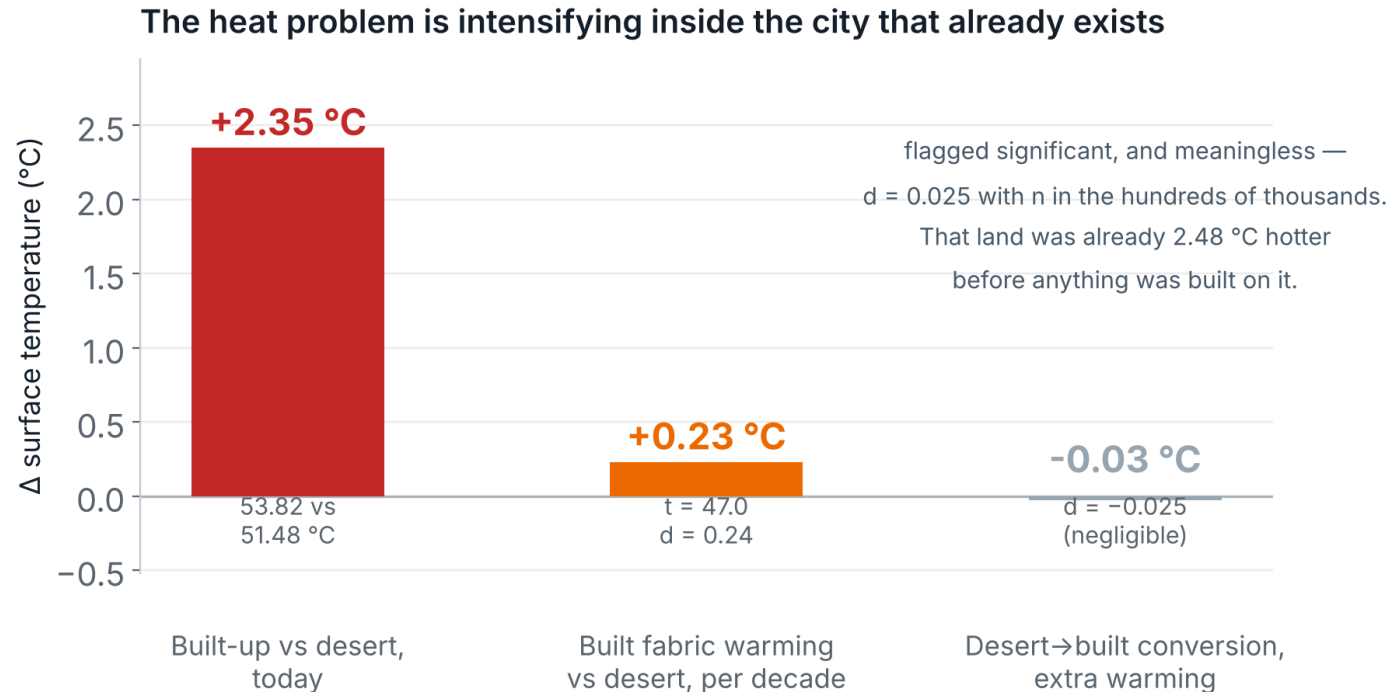
Repository

github.com/RISKYXBT/coolroof-mussafah

THE PROBLEM

Gulf cities are warming, and much of it is constructed

Replace bare desert with asphalt, concrete and dark roofing and albedo falls, shortwave absorption rises, evaporative cooling disappears, and stored heat re-radiates at night. A decade of Landsat over Mussafah measures all three effects.



The warming is concentrated in the fabric that **already exists**, not at the expansion frontier — which is what makes **retrofit**, rather than new-build regulation, the lever that matters now.

A municipality allocating a finite heat budget across thousands of buildings

Cool-roof coating is among the cheapest interventions available — but coating every roof in an industrial district is not affordable, and coating the wrong ones wastes the budget.

WHAT THEY USE TODAY

City-scale urban heat island maps from consultants, plus manual site survey. Those maps show **where it is hot**.

WHAT IS MISSING

Which roof to treat **first**. How much cooling the treatment **buys**. Whether that roof **can be coated at all**. The step from map to works programme is taken on judgement.

OWNERSHIP SPLITS THREE WAYS — AND THE MUNICIPALITY HAS A DIFFERENT LEVER FOR EACH		
WHO OWNS THE ROOF	MUNICIPALITY'S LEVER	WHAT THE RANKING IS USED FOR
ADM municipal buildings	Direct capital works	Sequencing its own retrofit programme against budget
Other government entities	Inter-entity coordination	Showing an entity its own stock, ranked, with the evidence
Private industrial landlords most of Mussafah's roof area	Regulation, permit conditions, incentives — not direct spending	Targeting where a code change buys the most cooling per unit of regulatory effort. A blanket requirement is politically expensive; one aimed at the 2.8% of roofs carrying 21% of the cooling is defensible .

A repeat screening study, not a software subscription

Procurement reality in Abu Dhabi favours consultancy studies on a multi-year cycle over recurring software. A three-to-four-year cadence also matches how fast the building stock and its surfaces actually change.

DELIVERABLE

Ranked roof inventory as **CSV + GIS layer**, heat and material maps, and a prioritisation report — scoped to a district or to a single entity's portfolio.

CADENCE

Every **3–4 years**, with the previous run as the baseline — so the second study also **measures what the first one achieved**.

COST BASE

Every input is **free and open**: Landsat, Sentinel-2, EMIT, OpenStreetMap. Essentially all cost is analyst time, which is why this is viable at district scale where a field survey of 3,603 roofs never would be.

BUYERS

ADM for its own stock. Other government entities for theirs. Large industrial landlords and free-zone operators for their portfolios.

On unit rates — deliberately left open. The pipeline outputs cooling per square metre and avoided absorption in kilowatts. It does **not** embed a coating price. ADM maintains its own schedule of rates; applying it to the `area_m2` column turns the ranking into a cost-effectiveness ranking in one multiplication. An invented AED/m² figure would be the first number a procurement reviewer challenged.

Four open sources, every scene ID recorded

Accessed through the Microsoft Planetary Computer STAC API, NASA Earthdata (CMR + LP DAAC) for EMIT, and the Overpass API for OpenStreetMap. Scene IDs, query windows and cloud thresholds are written to provenance .json on every run.

PRODUCT	PROVIDER	DATES	SCENES	RESOLUTION	LICENCE
Landsat 8/9 C2 L2SP	USGS / NASA	2014-05 → 2016-09	40 · cloud <15% · May–Sept only	30 m	Public domain
Landsat 8/9 C2 L2SP	USGS / NASA	2024-05 → 2026-09	40 · cloud <15% · May–Sept only	30 m	Public domain
Sentinel-2 L2A	ESA Copernicus	2022-09 → 2026-09	20 · cloud <15% · May–Sept only , 5 summers	10–20 m	Free and open
NASA EMIT L2A reflectance	NASA JPL / LP DAAC	2026-05-20, 15:03 local	1 granule	60 m · 285 bands	Open
ESA WorldCover v200	ESA	2021	tile N24E054	10 m	CC BY 4.0
OpenStreetMap buildings	OSM contributors	2026-10-07	8,503 raw footprints	vector	ODbL

WHY EMIT, AND WHY IT MATTERS

EMIT is an imaging spectrometer on the ISS whose science remit is **arid-region surface mineralogy** — which is exactly why **47 granules** cover Mussafah, far better coverage than curated commercial hyperspectral archives. It crosses at **15:03 local, three hours later than Landsat**, so it samples closer to peak surface temperature.

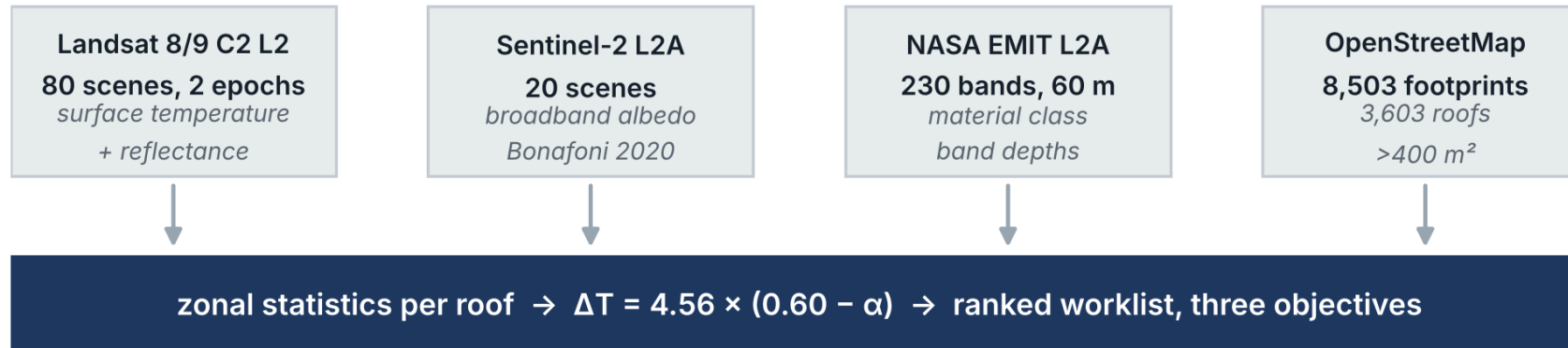
SINGLE-SENSOR CHANGE DETECTION

Both epochs use **Landsat only**. Mixing sensors across a decade introduces cross-calibration bias that is indistinguishable from the urban warming signal we are trying to measure.

APPROACH

Four open data streams into one ranked worklist

Two Jupyter notebooks, start to finish, on a free Colab runtime. No GPU, no paid API, no manual step.



every input free and open · no credentials · reproducible from the repo

STEP 1 — SURFACE TEMPERATURE

Landsat `lwir11`, scaled to kelvin, `qa_pixel` bit-masked for cloud, shadow, cirrus and fill. **May–Sept only**, per-pixel median.

STEP 2 — BUILT-UP MASK

Random forest on 14 features including **multi-scale texture**, required to agree with ESA WorldCover. Validated on held-out 2 km spatial blocks.

STEP 3 — ALBEDO AND SENSITIVITY

Sentinel-2 narrow-to-broadband albedo (Bonafoni & Şekertekin 2020, weights cross-checked against GRASS GIS `i.albedo`) from the **same months**, regressed against temperature under **spatial block cross-validation**.

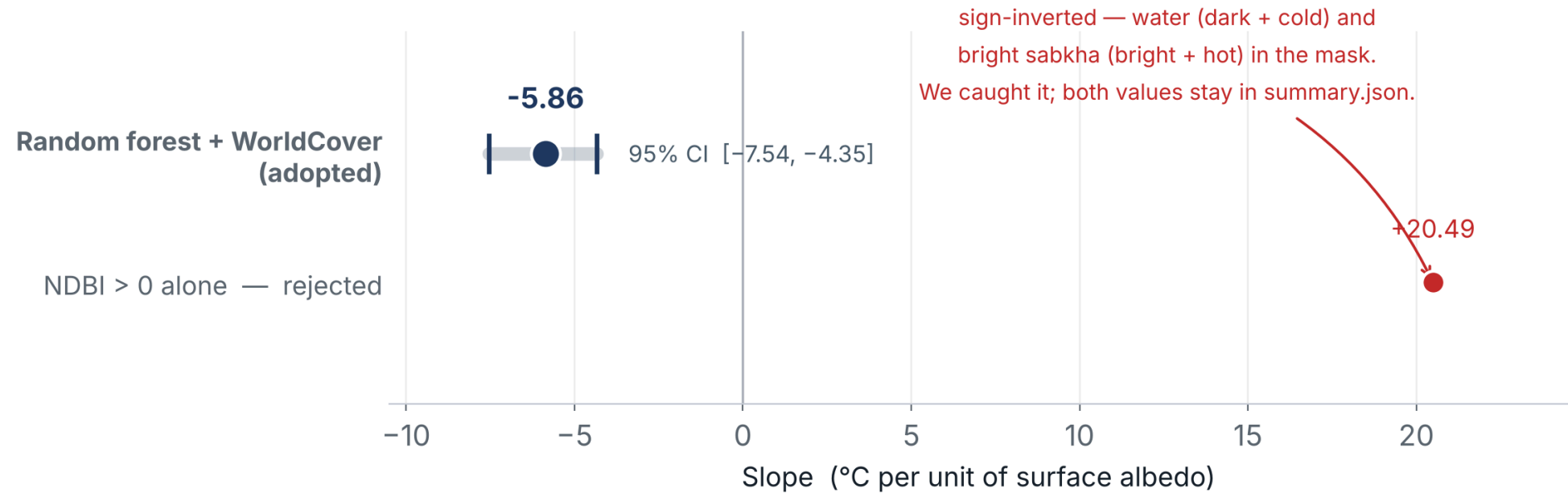
STEP 4 — THE WORKLIST

Zonal statistics per OSM footprint, $\Delta T = 5.86 \times (0.60 - \alpha)$, ranked under three objectives with EMIT material class attached.

THE CORE RESULT

We measured the lever instead of assuming it

Most cool-roof proposals quote a cooling figure from the literature — usually from a temperate city. We estimated it from Abu Dhabi's own summer surfaces, under spatial block cross-validation, against a deliberately contaminated mask as a control.



300 spatial block bootstraps · 175,033 px in 188 blocks · both masks are computed on every run and both values kept in summary.json

-5.86 °C

per unit of albedo, 95% CI [-7.54, -4.35]

0.59 °C

cooling per +0.10 albedo [0.44, 0.75]

175,033

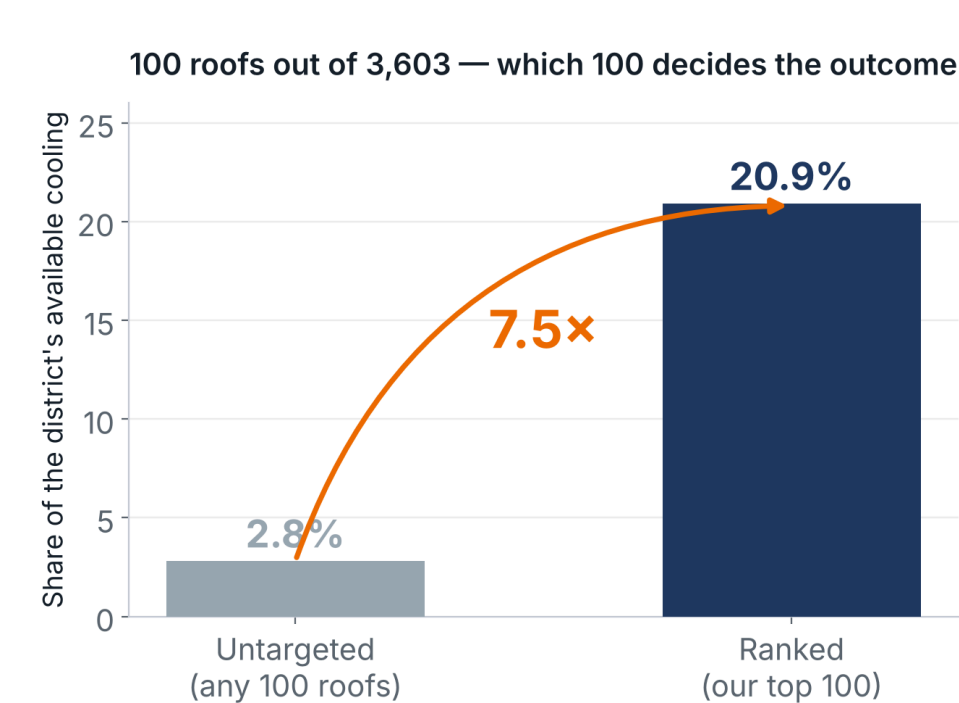
pixels in 188 spatial blocks

0.130

spatial-CV R² — and we report it plainly

3,603 roofs, ranked — and the objective is a policy choice

Zonal statistics over OpenStreetMap footprints larger than 400 m², each carrying albedo, surface temperature, area, predicted ΔT with bounds, and avoided solar absorption.



THREE RANKING MODES SHIP IN THE CSV

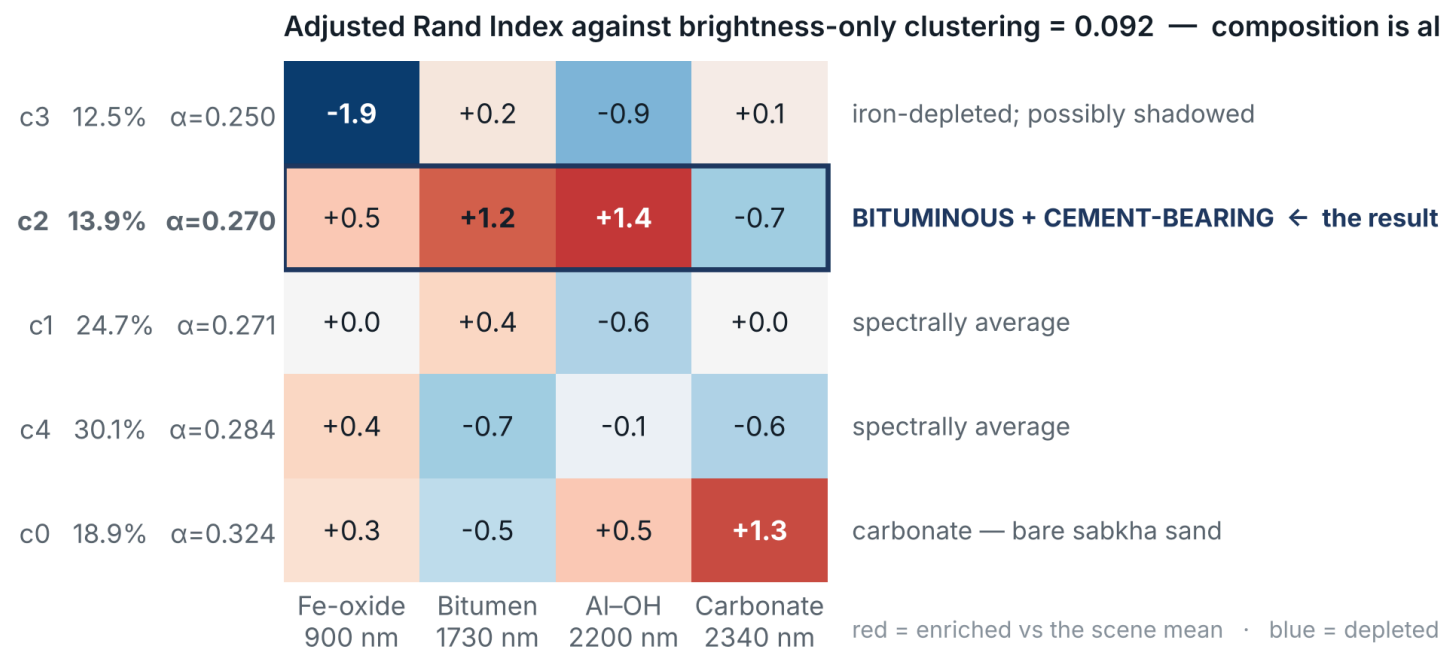
OBJECTIVE	TOP-100 AREA	MEAN LST	AVOIDED ABSORPTION
Total benefit area × cooling	99.4 ha	54.5 °C	56.5 MW
Hottest roofs first	48.1 ha	56.8 °C	22.3 MW
Biggest per-roof ΔT	5.0 ha	52.3 °C	5.4 MW

Only **21 of 100** roofs appear on both the "total benefit" and "hottest" lists. The choice of objective genuinely changes which buildings get treated — so it belongs to the municipality, not to us. All three ship.

Avoided absorption is **reduced shortwave load, not an air-conditioning saving**, at $S = 300 \text{ W/m}^2$ — a round value 3.4% above the **290 W/m² highest monthly mean measured in Abu Dhabi** (Islam et al. 2009, a full year of ground pyranometer records). A 24-hour mean, so this is averaged over the whole diurnal cycle; at the measured 290 it is 54.6 MW.

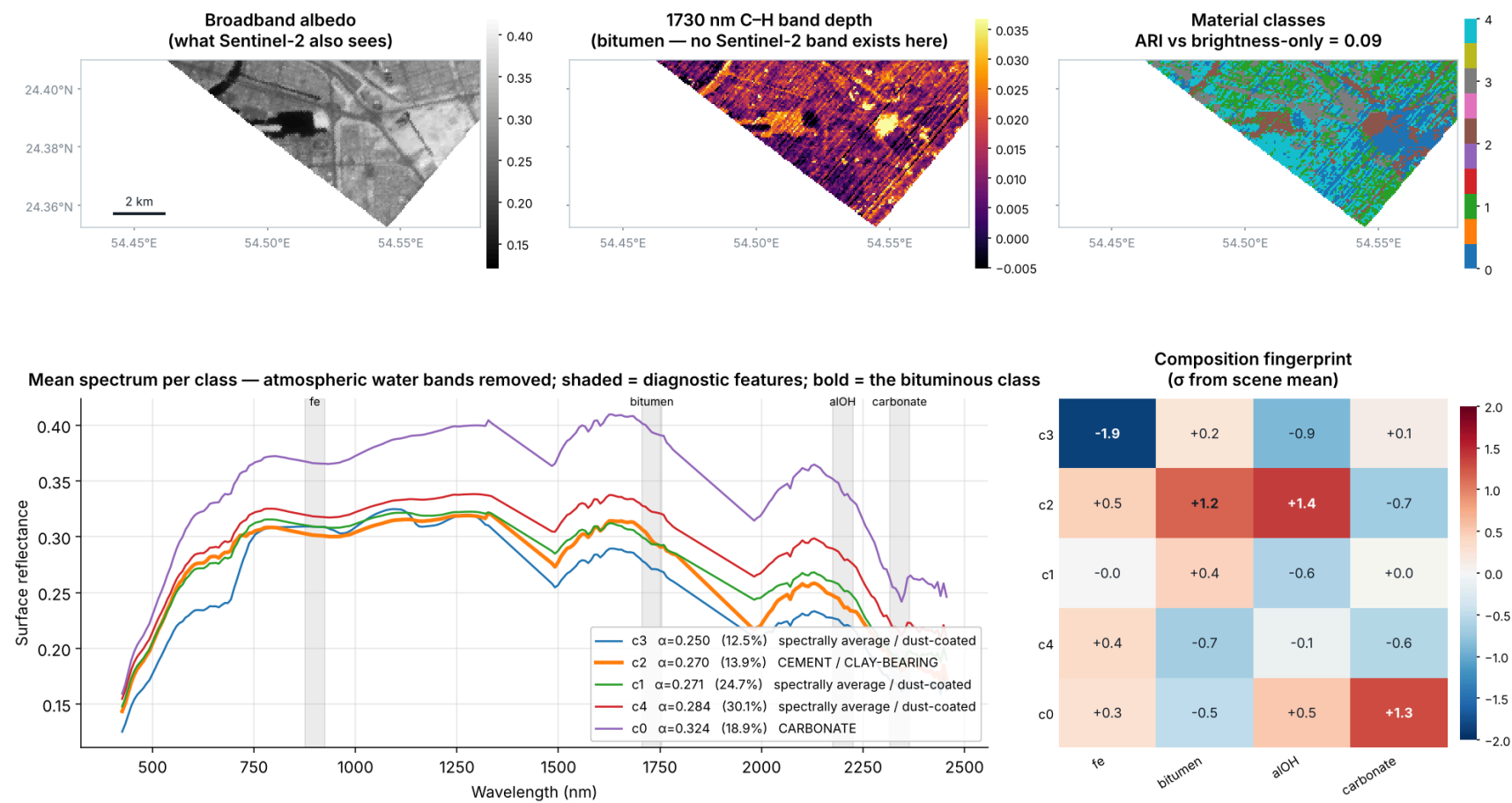
Albedo says a roof is dark. Only narrow bands say what it is made of.

Band depths against a local two-point continuum (Clark & Roush 1984) at four diagnostic features, then k-means on the depths standardised to the scene mean — so classes separate on **composition**, not brightness.



There is **no Sentinel-2 band at 1730 nm**. The C–H enrichment that separates a bituminous roof — which can be coated — from a dark metal roof that has to be re-covered is only available from a spectrometer. Those are different line items on a schedule of rates, by roughly an order of magnitude.

EMIT over Mussafah — real hyperspectral, real study area



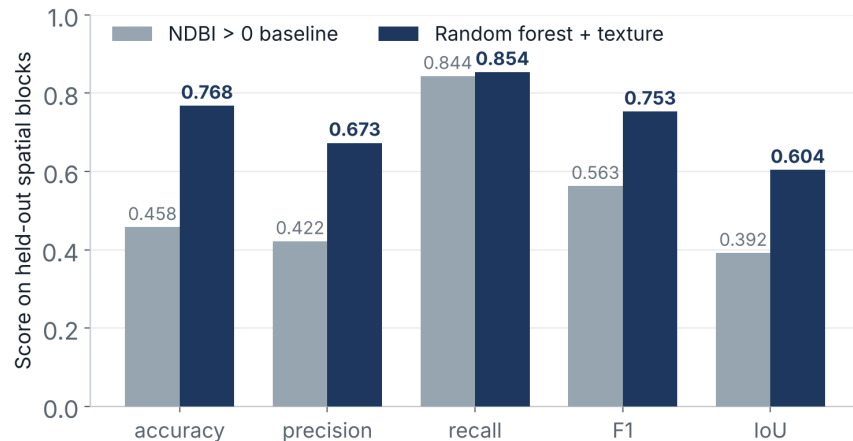
EMIT granule EMIT_L2A_RFL_001_20260520T110418_2614007_042, 2026-05-20 15:04 local · maps drawn at true geographic aspect (15.2 × 6.4 km) · the wedge is the EMIT swath corner clipping the AOI, 43.5% GLT fill

Left to right: broadband albedo (what Sentinel-2 also sees) · 1730 nm C-H band depth (bitumen — only hyperspectral sees this) · material classes, ARI 0.09 against brightness-only clustering. Below: mean spectrum per class with the diagnostic features shaded, and the composition fingerprint in σ from the scene mean.

results/10_emit_final.png — produced by notebooks/04_emit_hyperspectral.ipynb

Held-out spatial blocks, and the mistakes we caught

Never random pixel splits. Adjacent pixels in satellite imagery are not independent, so a random split leaks the test set into training and inflates every metric.



2 km blocks · 60/40 split · $n = 95,346$ test pixels · recall is unchanged, so the whole gain is in false positives: $45,533 \rightarrow 16,345$

A SIGN ERROR WE CAUGHT AND FIXED

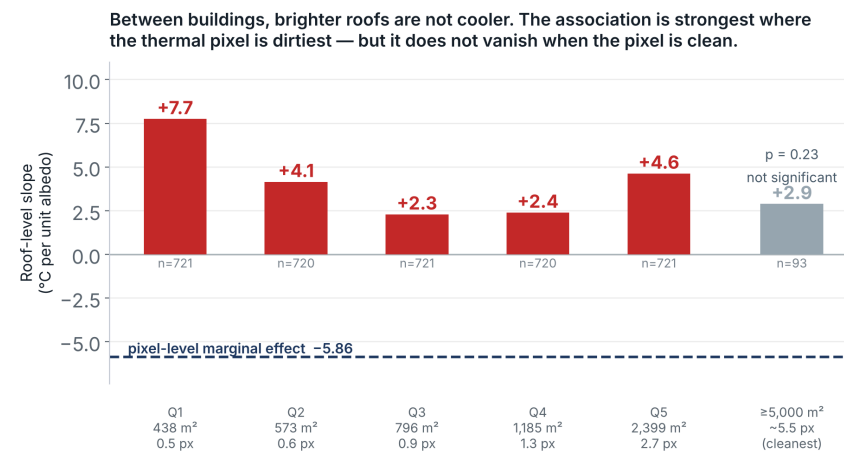
A loose mask ($\text{NDBI} > 0$ alone) gives **+20.49** — brighter surfaces appearing hotter, inverting the physics. Cause: channel water (dark **and** cold) anchoring the low end, bright sabkha (bright **and** very hot) anchoring the high end. Requiring agreement with an independent land-cover product recovered the expected negative slope. The notebook computes **both masks on every run** and keeps both values in summary.json — the audit trail is part of the evidence.

ON THE R^2 (0.130)

We estimate a **marginal effect, not a prediction**. Absolute surface temperature is driven mostly by factors we do not model — thermal mass, moisture, building use, HVAC exhaust. The coefficient is nonetheless tightly constrained: its entire 95% interval lies below zero. Matching the albedo and temperature seasons **more than doubled R^2 , from 0.057** — the strongest evidence that the season filter was the right call.

The roof cross-section does not confirm the per-roof predictions

We ranked the roofs, then tested the ranking against itself: across the 3,603 buildings, is a brighter roof a cooler roof? **No** — the raw association is **+6.6 °C per unit albedo** ($p \approx 10^{-75}$) and survives controlling for roof size (+4.8). We report it because a reviewer who opens our own CSV will find it.



PART OF IT IS THE THERMAL PIXEL

Landsat thermal is 30 m; the median roof here is **795 m², under one pixel**, so its "roof temperature" is largely its surroundings. Strongest for sub-pixel roofs (+7.7), falling to **+2.9, not significant**, for the 93 roofs spanning ~5 clean pixels.

THE REST IS A BETWEEN-BUILDING CONFOUND

Brighter-roofed buildings here are **different buildings**: larger, newer, in wider paved yards, with different internal heat loads and roof thermal mass. None of that is in our data, and looking across buildings does not remove it.

WHY THIS DOES NOT REFUTE -5.86

Two different questions. **"Do brighter surfaces run cooler, location and vegetation fixed?"** — yes, -5.86 within spatial blocks. **"Are buildings with brighter roofs cooler buildings?"** — no, which is a fact about which buildings have bright roofs, not about radiation.

WHAT IT COSTS US — STATED PLAINLY

The per-roof ΔT column is a **screening estimate carried down from a pixel-level marginal effect, not validated at roof scale**. Settling it needs a before/after on coated roofs — step 1 of the next phase.

What this does not do

A NULL RESULT WE REPORT AS ONE

Land converted from desert to built-up **did not warm more** than desert left untouched: $-0.03\text{ }^{\circ}\text{C}$, flagged significant at $p = 10^{-5}$ but with **Cohen's d = 0.025** — with n in the hundreds of thousands a p-value detects any departure from zero, and this one is a twenty-fifth of a standard deviation. Before any construction that land was already **$2.48\text{ }^{\circ}\text{C}$ hotter** than desert that stayed desert. Two explanations fit — **siting** (development hugs existing urban fabric) and **site preparation** (clearing removes the salt crust and shallow moisture of natural sabkha) — and **we cannot separate them** with the data we have.

WHAT HYPERSPECTRAL DOES NOT DO HERE

Three of five classes — **67% of the imaged area** — are spectrally average, no feature more than 0.7σ from the scene mean. Aeolian dust homogenises surface spectra in an arid industrial zone, so hyperspectral meaningfully differentiates roughly **one third** of the scene. Any operational hyperspectral urban product over Gulf cities, **Satellite 813 included**, will face the same constraint.

- **The per-roof ΔT is not validated at roof scale** — see the previous slide. It is a screening estimate, not an engineering prediction.
- **The albedo composite is September-weighted** within the summer — 17 of 20 Sentinel-2 scenes. Far better than the cool-season composite it replaced, but still not sampled identically to temperature.
- **Albedo spans five summers, temperature three.** A roof re-covered in 2023 contributes its old surface to the albedo median.
- Land surface temperature is **not air temperature**. This ranks surfaces; it does not predict heat-stroke risk.
- The sensitivity is **correlational**. Roof thermal mass, building use and HVAC exhaust are not separated.

Full list in README §9.7 — each stated before a reviewer has to find it

- Validation is against **ESA WorldCover, itself a model**. No field survey was performed.
- The classifier **over-predicts built-up** (52.4% vs 41.3% reference); precision 0.673.
- EMIT coverage is **partial**: $\sim 46\text{ km}^2$ of the 195 km^2 AOI, the northern strip, at 43.5% GLT fill.
- Both band-depth maps show **detector striping**, so part of those signals is instrumental.
- Material classes are **"consistent with"** a material, never confirmed. No field verification.
- OSM footprint completeness varies. A deployed version would use **municipal cadastral data**.

What exists today, and what the next phase needs

3,603

roofs ranked — 4.82 km² of roof area

7.5×

benefit concentration in the top 100 roofs

56.5 MW

avoided solar absorption, summer mean

AED 0

data cost — every input free and open

1 · GROUND TRUTH AND A COATED-ROOF TRIAL

A rooftop campaign on 20–30 roofs across the five spectral classes: verifies the material call, gives in-situ albedo, and — by coating a subset and re-imaging — **measures ΔT at roof scale directly**, which is the one claim our own data cannot confirm.

2 · CADASTRAL FOOTPRINTS

Replace OpenStreetMap with ADM's own building layer. Removes the completeness gap and attaches **ownership** to every ranked roof — which is what routes each one to the right municipal lever.

3 · RESOLVE THE NULL

A matched-control design pairing converted parcels with equidistant unconverted ones, plus construction-timing records, to separate **siting** from **site preparation**. Needed before anyone cites a conversion-warming figure.

4 · RE-RUN AS THE BASELINE

The pipeline is the study and the study is repeatable. Run it again in 3–4 years and the same code **measures what the first programme achieved**.

A reviewer who has never spoken to us can **clone the repository, follow the README, and reproduce every number on these slides** — two notebooks, free data, no credentials.

Team T0050

MEMBER	ROLE
Meera Almansoori	Team lead · surface energy balance physics, methodology, validation design
Ghazi Hafedh	Data pipeline, modelling, software, repository
Ahmed Abu Farha	Municipal building-control domain input, end-user requirements

DATA ATTRIBUTION

Landsat Collection 2 Level-2 courtesy of the U.S. Geological Survey.
Contains modified Copernicus Sentinel-2 data (2022–2026).
EMIT data courtesy of NASA JPL, distributed by the LP DAAC.
© ESA WorldCover project 2021, CC BY 4.0.
Map data © OpenStreetMap contributors, Open Database Licence.

METHODS CITED

Bonafoni, S. & Şekertekin, A. (2020). *Albedo Retrieval From Sentinel-2 by New Narrow-to-Broadband Conversion Coefficients*. IEEE Geoscience and Remote Sensing Letters, 17(9), 1618–1622.
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Islam, M. D., Kubo, I., Ohadi, M. & Alili, A. A. (2009). *Measurement of solar energy radiation in Abu Dhabi, UAE*. Applied Energy, 86(4), 511–515.
doi:10.1016/j.apenergy.2008.07.012

Clark, R. N. & Roush, T. L. (1984). *Reflectance spectroscopy: quantitative analysis techniques for remote sensing applications*. JGR Solid Earth.

USGS (2024). *Landsat Collection 2 Level-2 Science Product Guide*.

BUILT ON

Starter notebooks and data-access patterns from the official 813 Challenge repository by Dr. Vincent Markiet, Space42.

Code licence: MIT.

No credentials, API keys or restricted imagery are committed to the repository.